Electrical Circuit (1)

Introduction (week3 class1)

Dr. Akram Al-Mahrouk

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Chapter 3 Methods of Analysis 81

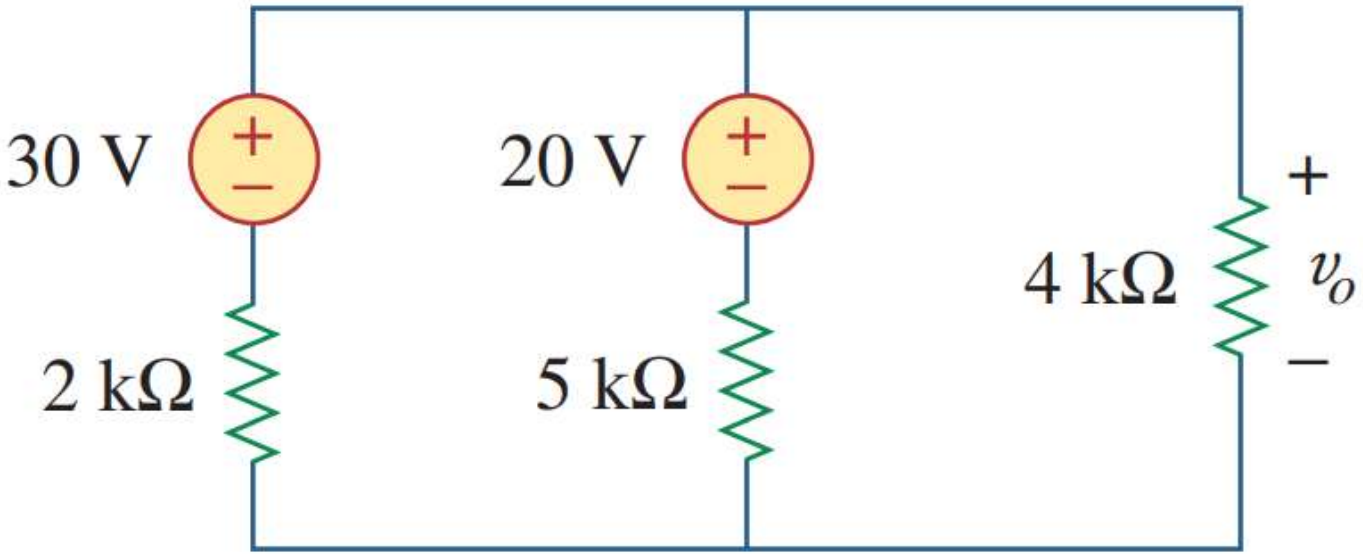
3.1 Introduction 82
3.2 Nodal Analysis 82
3.3 Nodal Analysis with Voltage Sources 88
3.4 Mesh Analysis 93
3.5 Mesh Analysis with Current Sources 98
3.6 †Nodal and Mesh Analyses by Inspection 100
3.7 Nodal Versus Mesh Analysis 104
3.8 Circuit Analysis with *PSpice* 105
3.9 †Applications: DC Transistor Circuits 107
3.10 Summary 112
Review Questions 113
Problems 114
Comprehensive Problem 126

Nodal Analysis

Steps to Determine Node Voltages:

1. Select a node as the reference node. Assign voltages v_1, $v_2, \ldots, v_{n-1}$ to the remaining $n - 1$ nodes. The voltages are referenced with respect to the reference node.
2. Apply KCL to each of the $n - 1$ nonreference nodes. Use Ohm's law to express the branch currents in terms of node voltages.
3. Solve the resulting simultaneous equations to obtain the unknown node voltages.

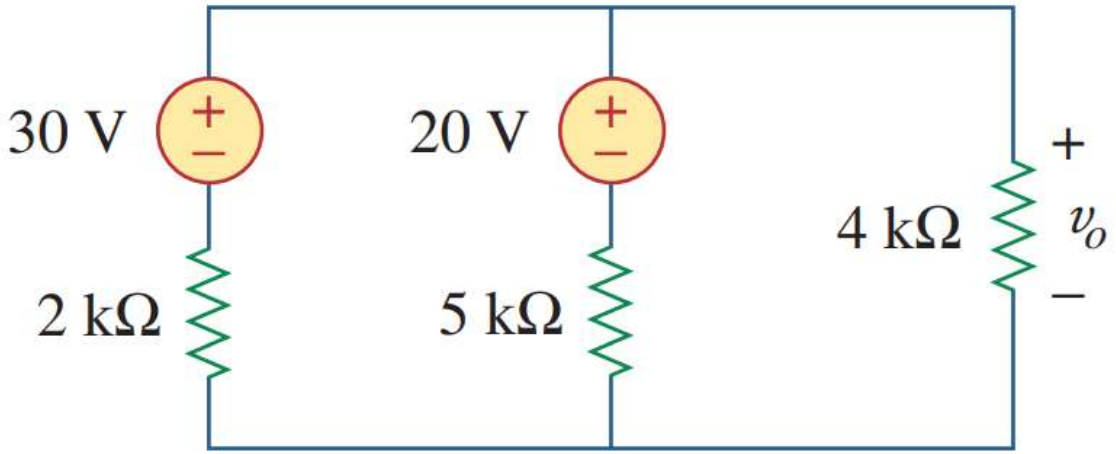
Nodal Analysis



Nodal Analysis

Basic equation

$$\frac{V_o - 30}{2000} + \frac{V_o - 20}{5000} + \frac{V_o}{4000} = 0$$



Formulated equation

$$\frac{V_o}{2000} + \frac{V_o}{5000} + \frac{V_o}{4000} = \frac{30}{2000} + \frac{20}{5000}$$

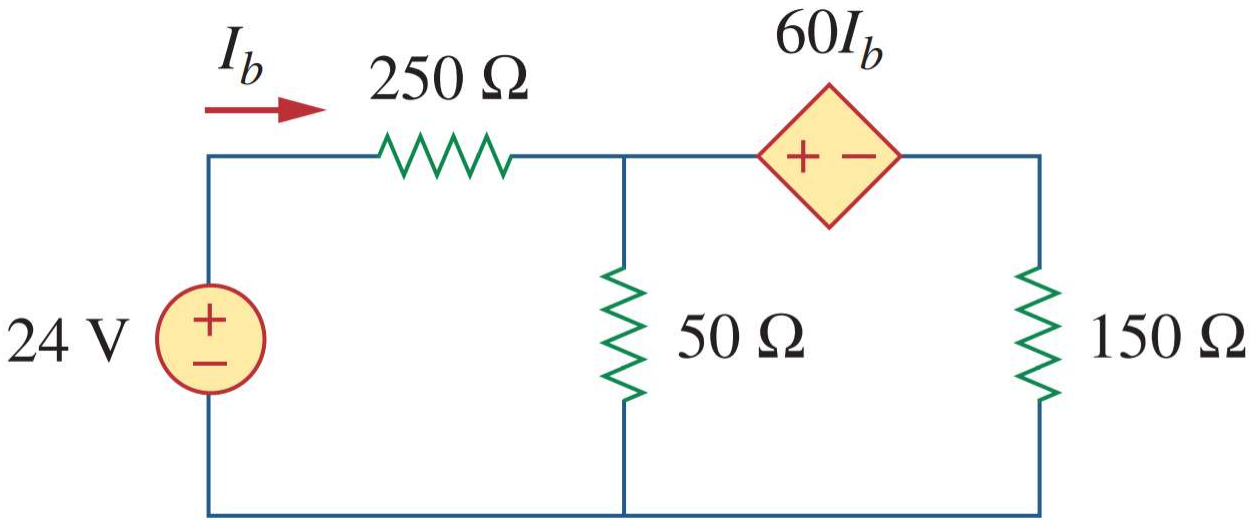
Final answer

$$V_o = 20\,V$$

Additional answer

$$XXXXXX$$

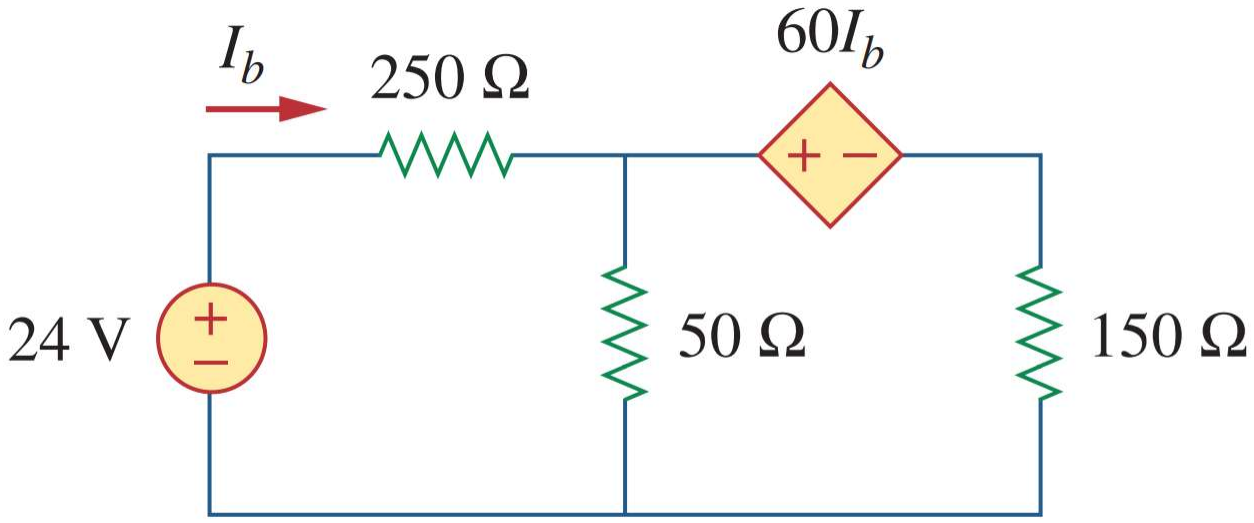
Nodal Analysis



Nodal Analysis

Basic equation

$$\frac{V_o - 24}{250} + \frac{V_o}{50} + \frac{V_o - 60(\frac{24 - V_o}{250})}{150} = 0$$



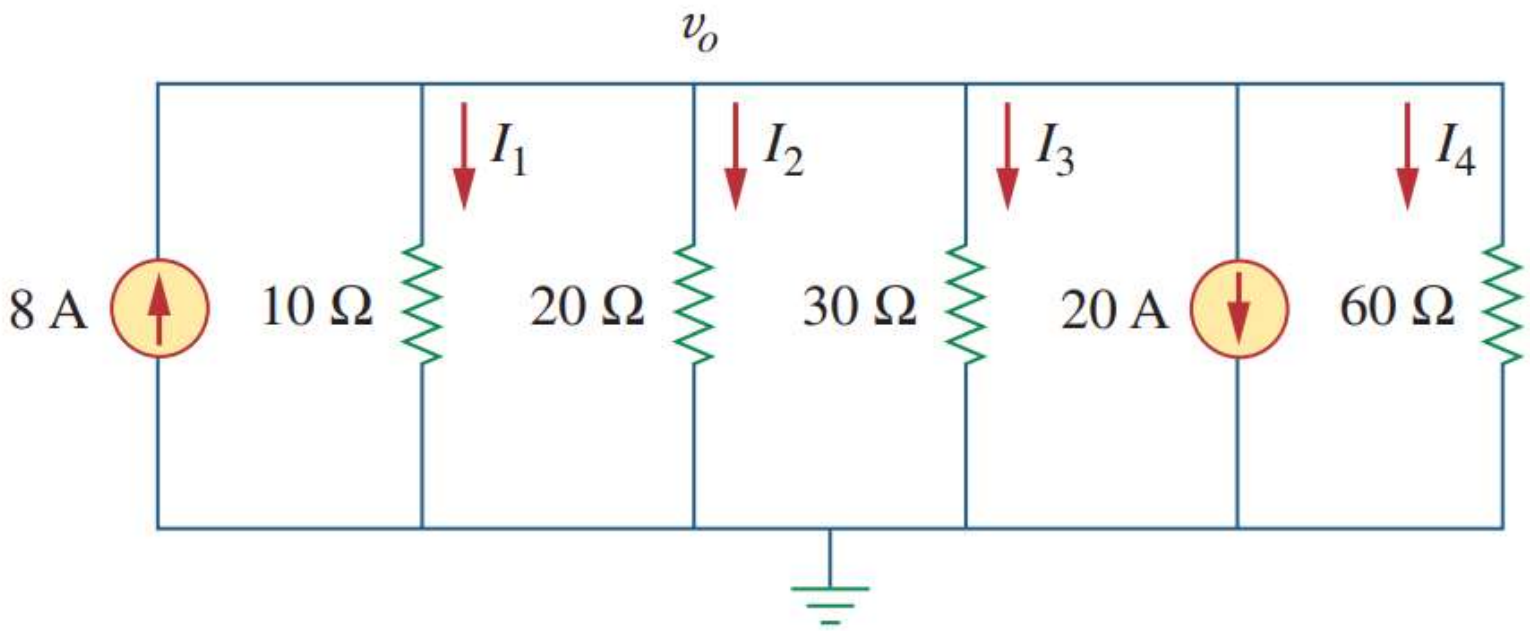
Formulated equation

$$\frac{V_o}{250} + \frac{V_o}{50} + \frac{V_o}{150} + \frac{(60 * V_o)/250}{150} = \frac{24}{250} + \frac{(60 * 24)/250}{150}$$

Final answer

$$V_o = 4.16\,V$$

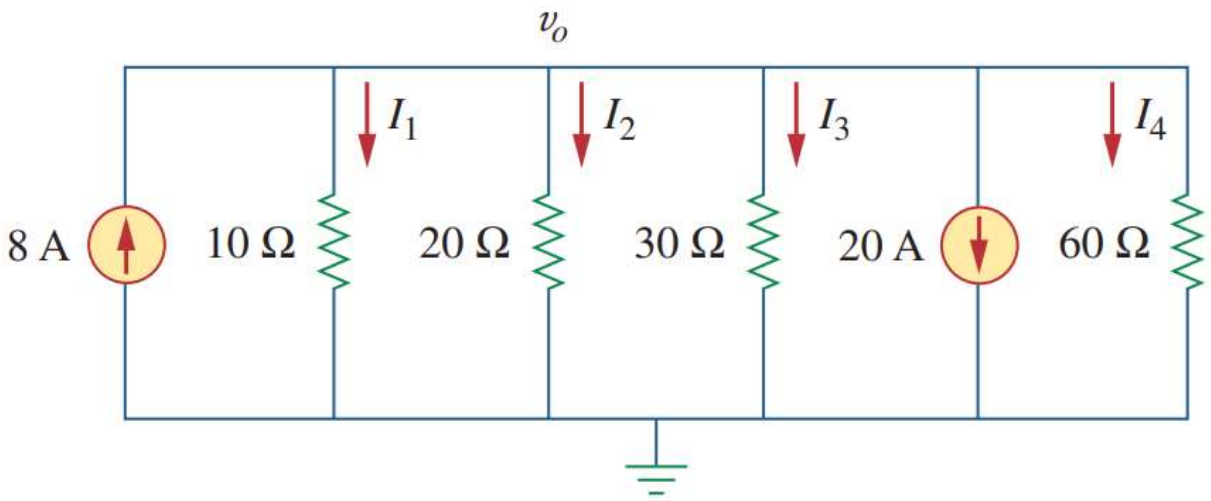
Nodal Analysis



Nodal Analysis

Basic equation

$$-8 + \frac{V_o}{10} + \frac{V_o}{20} + \frac{V_o}{30} + 20 - \frac{V_o}{60} = 0$$



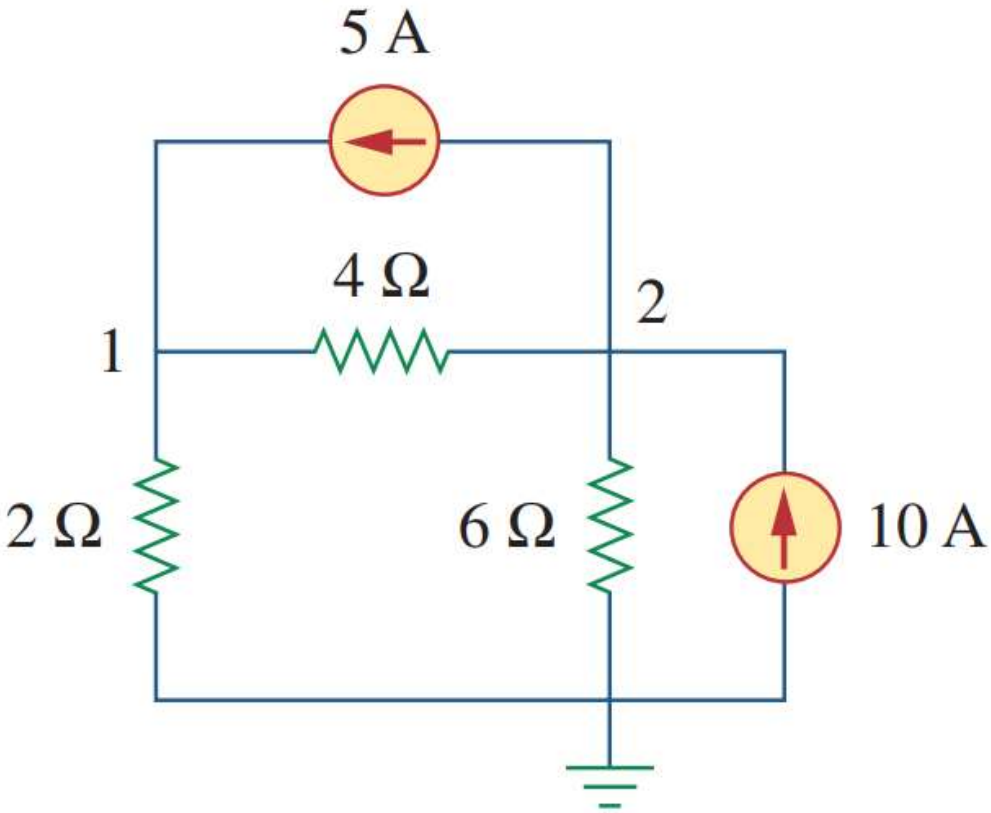
Formulated equation

$$\frac{V_o}{10} + \frac{V_o}{20} + \frac{V_o}{30} + \frac{V}{60} = -20 + 8$$

Final answer

$$V_o = -60\ V$$

Nodal Analysis



Nodal Analysis

Basic equation

$$\frac{V_1}{2} + \frac{V_1 - V_2}{4} - 5 = 0$$

$$\frac{V_2}{6} - 10 + \frac{V_2 - V_1}{4} + 5 = 0$$

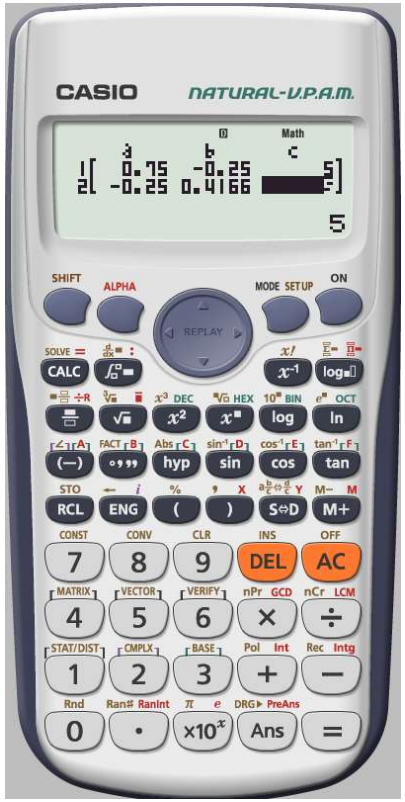
Formulated equation

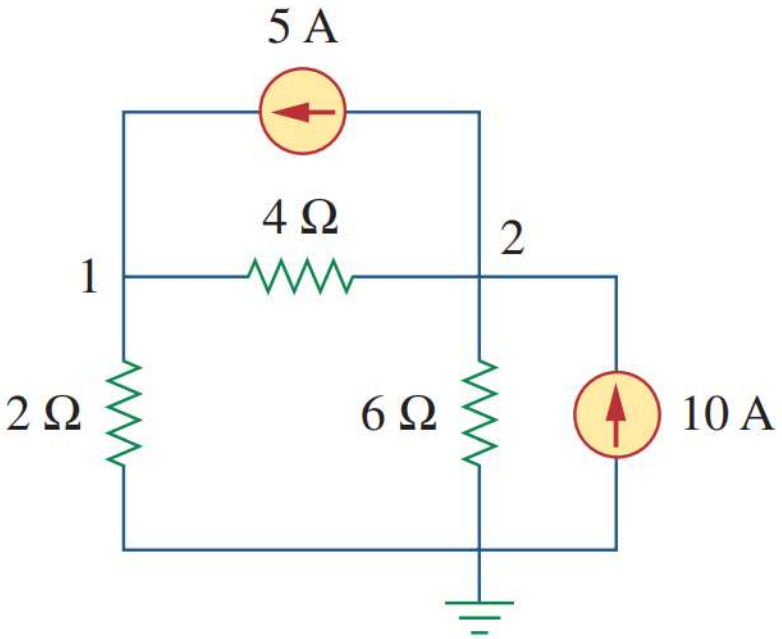
$$\left(\frac{1}{2} + \frac{1}{4}\right)V_1 - \frac{1}{4}V_2 = 5$$

$$\left(\frac{-1}{4}\right)V_1 + \left(\frac{1}{6} + \frac{1}{4}\right)V_2 = 5$$

Final answer

$V_1 = 13.333\ V$ $V_2 = 20\ V$







Electrical Circuit (1)

Introduction (week3 class2)

Dr. Akram Al-Mahrouk

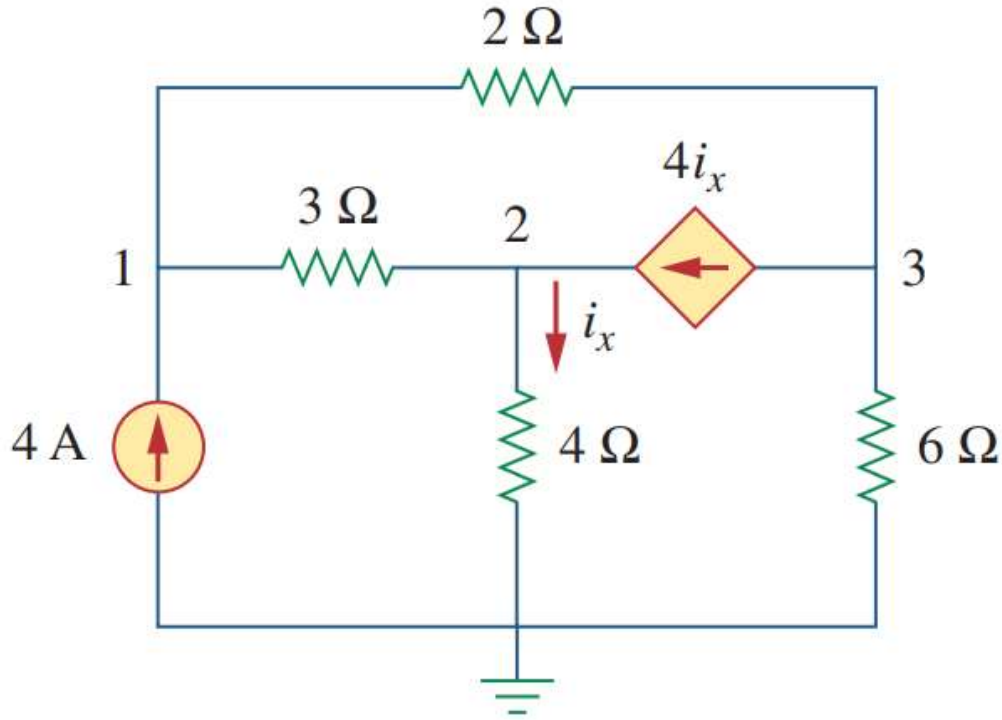
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Chapter 3 Methods of Analysis 81

3.1 Introduction 82
3.2 Nodal Analysis 82
3.3 Nodal Analysis with Voltage Sources 88
3.4 Mesh Analysis 93
3.5 Mesh Analysis with Current Sources 98
3.6 †Nodal and Mesh Analyses by Inspection 100
3.7 Nodal Versus Mesh Analysis 104
3.8 Circuit Analysis with *PSpice* 105
3.9 †Applications: DC Transistor Circuits 107
3.10 Summary 112
Review Questions 113
Problems 114
Comprehensive Problem 126

Advanced

Nodal Analysis



Nodal Analysis

Basic equation

$$-4+\frac{V_1-V_2}{3}+\frac{V_1-V_3}{2}=0$$

$$\frac{V_2-V_1}{3}+\frac{V_2}{4}-4\frac{V_2}{4}=0$$

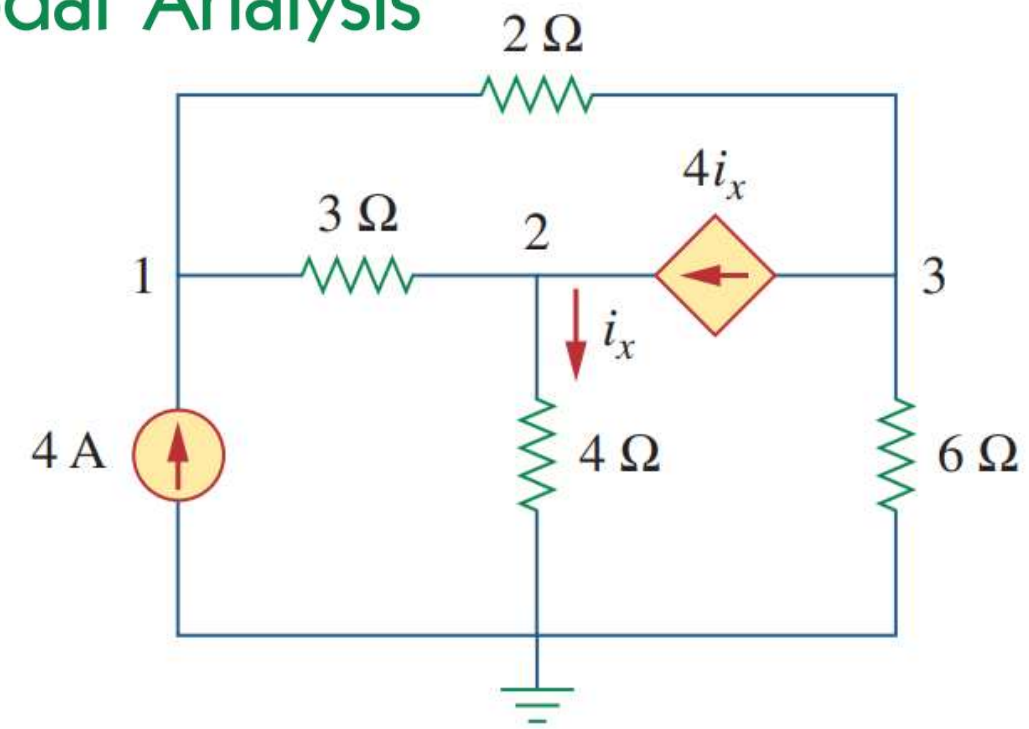
$$\frac{V_3-V_1}{2}+\frac{V_3}{6}+4\frac{V_2}{4}=0$$

Formulated equation

$$\left(\frac{1}{3}+\frac{1}{2}\right)V_1-\left(\frac{1}{3}\right)V_2-\left(\frac{1}{2}\right)V_3=4$$

$$\left(\frac{-1}{3}\right)V_1+\left(\frac{1}{3}+\frac{1}{4}-\frac{4}{4}\right)V_2=0$$

$$\left(\frac{-1}{2}\right)V_1+\left(\frac{4}{4}\right)V_2+\left(\frac{1}{2}+\frac{1}{6}\right)V_3=0$$



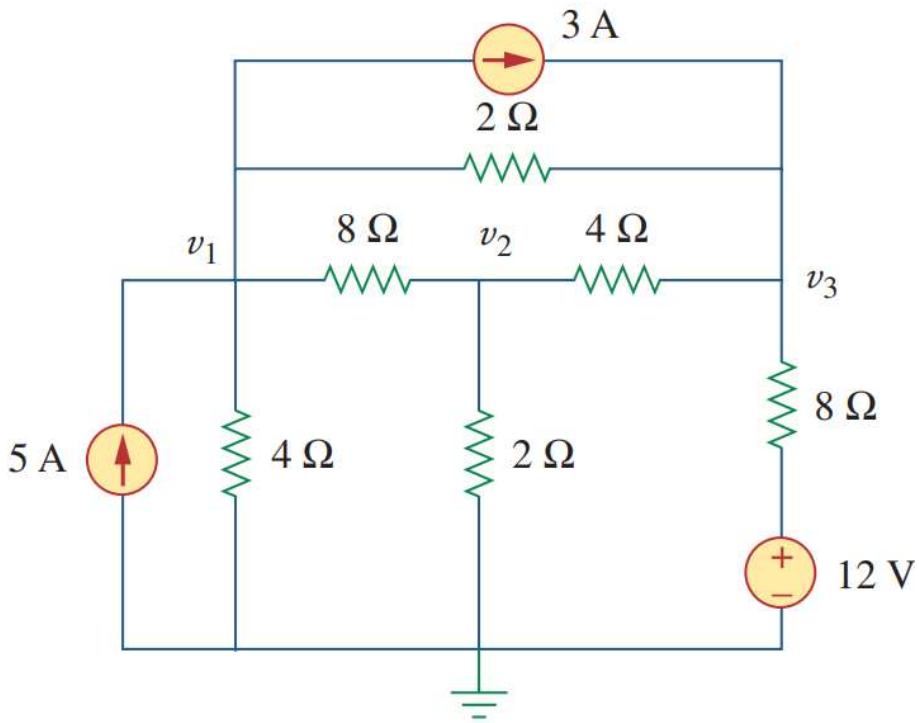
Final answer

$$V_1 = 32\ V$$

$$V_2 = -25.6\ V$$

$$V_3 = 62.4\ V$$

Nodal Analysis



Nodal Analysis

Basic equation

$$-5+\frac{V_1}{4}+\frac{V_1-V_2}{8}+\frac{V_1-V_3}{2}+3=0$$

$$\frac{V_2-V_1}{8}+\frac{V_2-V_3}{4}+\frac{V_2}{2}=0$$

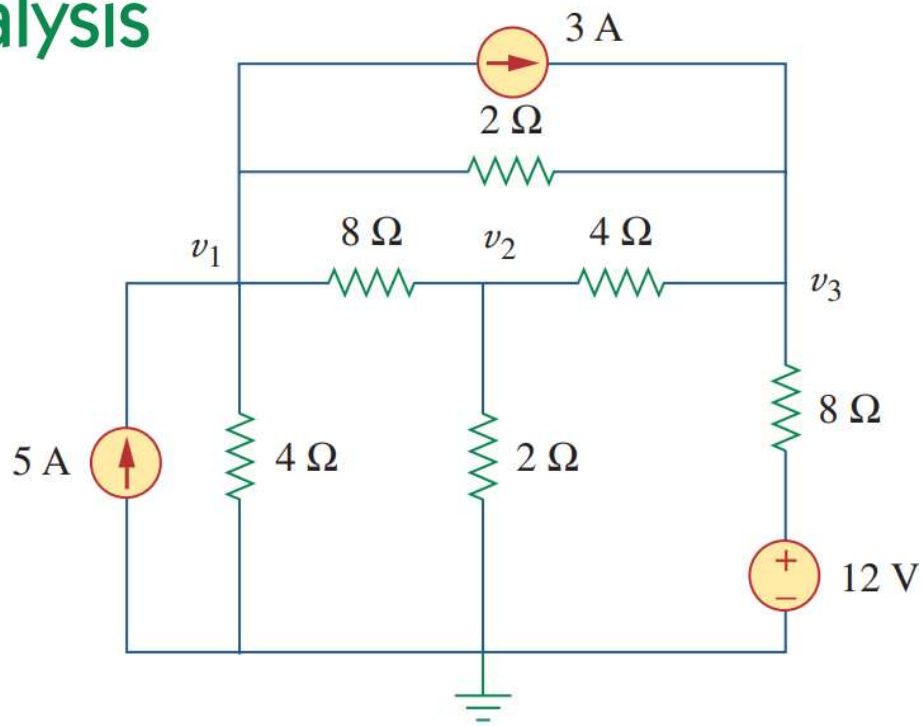
$$-3+\frac{V_3-V_1}{2}+\frac{V_3-V_2}{4}+\frac{V_3-12}{8}=0$$

Formulated equation

$$\left(\frac{1}{4}+\frac{1}{8}+\frac{1}{2}\right)V_1-\left(\frac{1}{8}\right)V_2-\left(\frac{1}{2}\right)V_3=5-3$$

$$\left(\frac{-1}{8}\right)V_1+\left(\frac{1}{8}+\frac{1}{4}+\frac{1}{2}\right)V_2+\left(\frac{-1}{4}\right)V_1=0$$

$$\left(\frac{-1}{2}\right)V_1+\left(-\frac{1}{4}\right)V_2+\left(\frac{1}{2}+\frac{1}{4}+\frac{1}{8}\right)V_3=3+\frac{12}{8}$$



Final answer

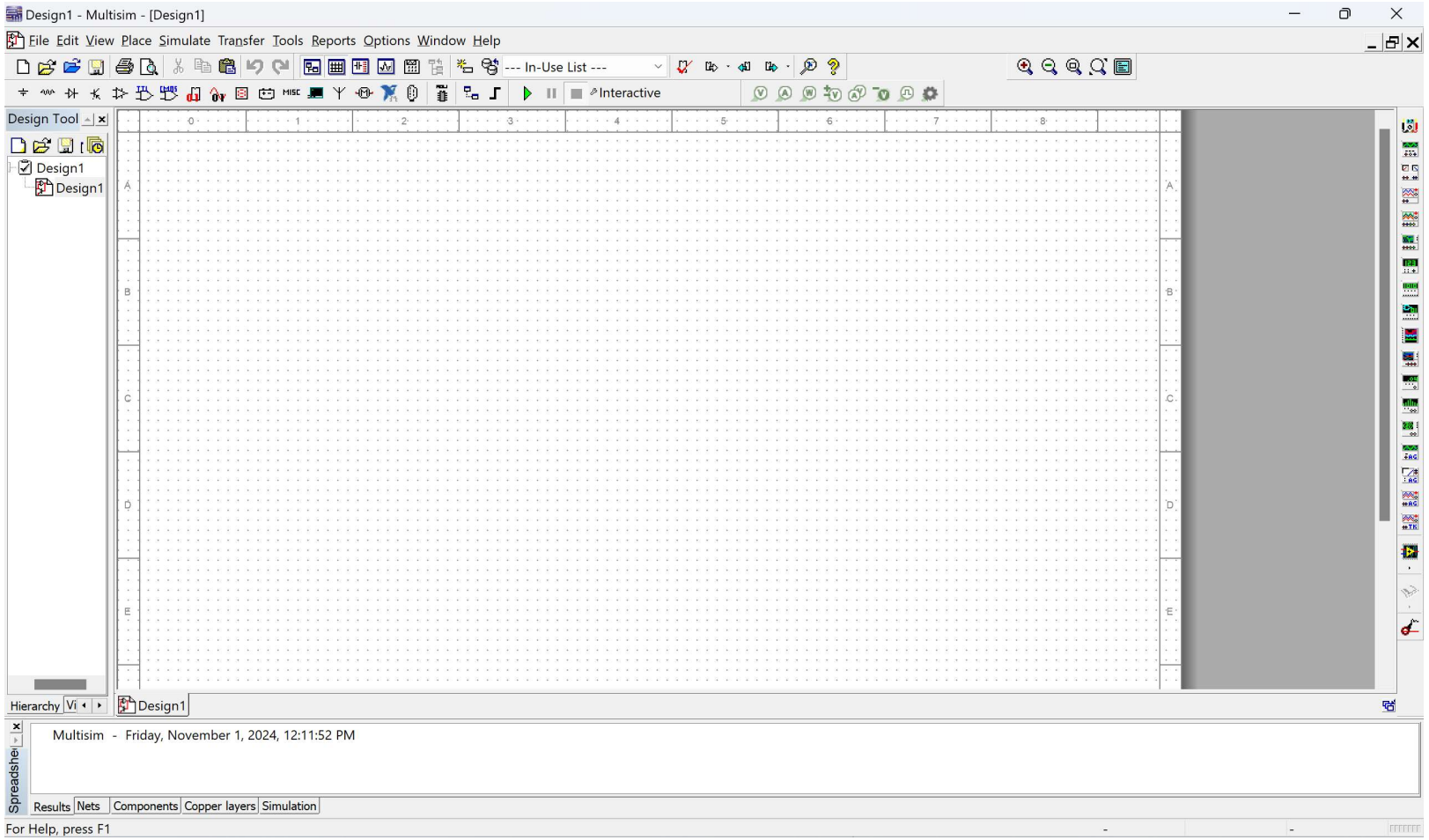
$$V_1 = 10\ V$$

$$V_2 = 4.9333\ V$$

$$V_3 = 12.2666\ V$$

NATIONAL INSTRUMENTS







Electrical Circuit (1)

Introduction (week4 class1)

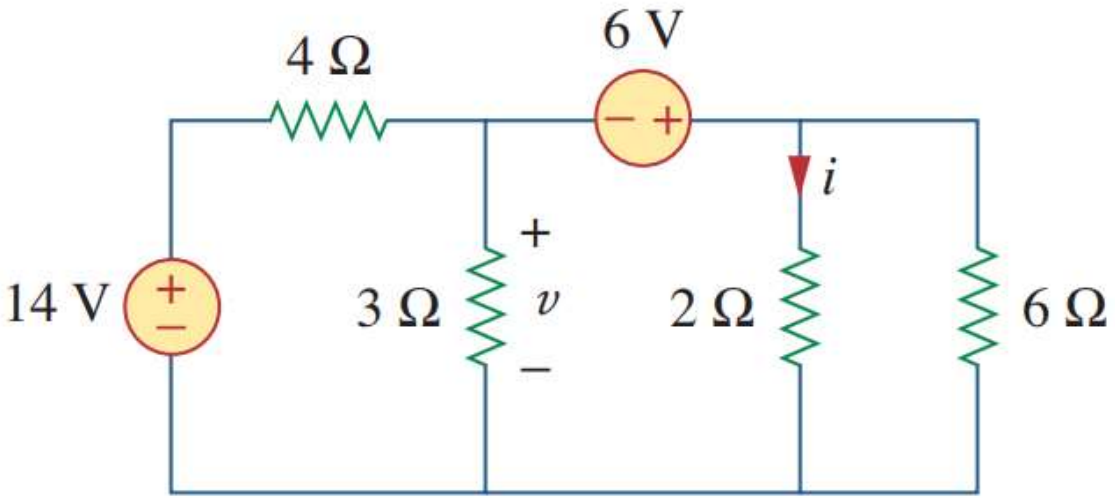
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Chapter 3 Methods of Analysis 81

3.1 Introduction 82
3.2 Nodal Analysis 82
3.3 Nodal Analysis with Voltage Sources 88
3.4 Mesh Analysis 93
3.5 Mesh Analysis with Current Sources 98
3.6 †Nodal and Mesh Analyses by Inspection 100
3.7 Nodal Versus Mesh Analysis 104
3.8 Circuit Analysis with *PSpice* 105
3.9 †Applications: DC Transistor Circuits 107
3.10 Summary 112
Review Questions 113
Problems 114
Comprehensive Problem 126

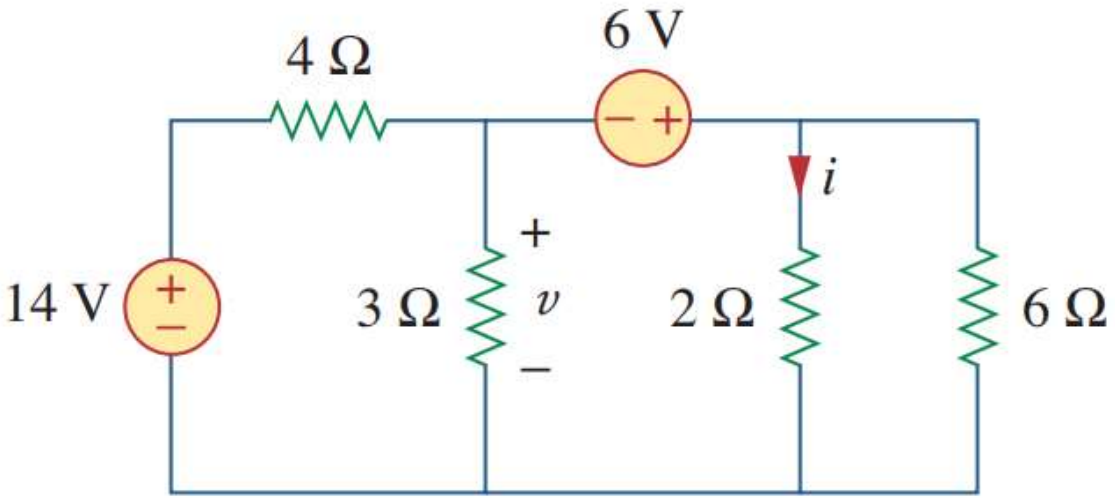
Nodal Analysis with Voltage Sources





CalcES

Nodal Analysis with Voltage Sources



Basic equation

$$V_1 + 6 = V_2$$

$$\frac{V_1 - 14}{4} + \frac{V_1}{3} + \frac{V_2}{2} + \frac{V_2}{6} = 0$$

Formulated equation

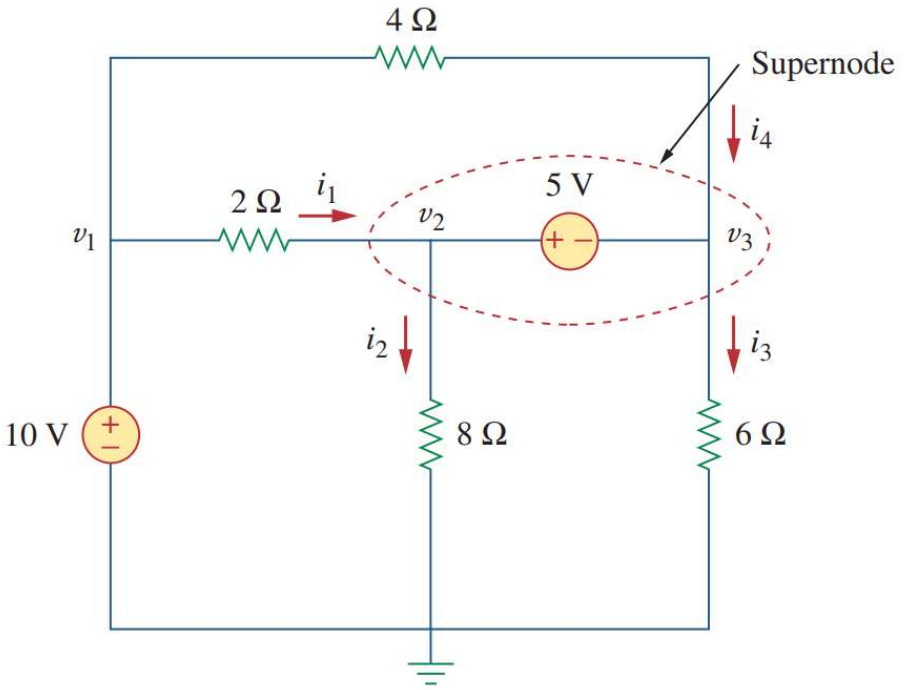
$$V_1 - V_2 = -6$$

$$\left(\frac{1}{4} + \frac{1}{3}\right)V_1 + \left(\frac{1}{2} + \frac{1}{6}\right)V_2 = \frac{14}{4}$$

Final answer

$$V_1 = -0.4\,mV \qquad V_2 = 5.6\,V$$

Nodal Analysis with Voltage Sources



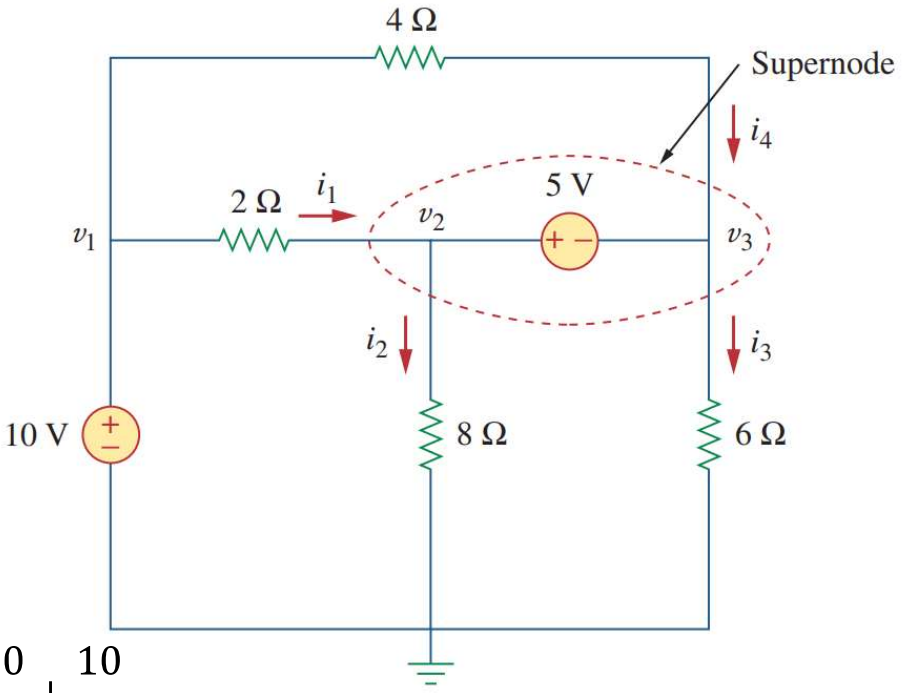
Nodal Analysis with Voltage Sources

Basic equation

$V_1 = 10$

$V_3 + 5 = V_2$

$$\frac{V_2 - V_1}{2} + \frac{V_2}{8} + \frac{V_3 - V_1}{4} + \frac{V_3}{6} = 0$$



Formulated equation

$V_2 - V_3 = 5$

$$\frac{V_2 - 10}{2} + \frac{V_2}{8} + \frac{V_3 - 10}{4} + \frac{V_3}{6} = 0 \qquad \left(\frac{1}{2} + \frac{1}{8}\right)V_2 + \left(\frac{1}{4} + \frac{1}{6}\right)V_3 = \frac{10}{2} + \frac{10}{4}$$

Final answer

$V_1 = 10\ V$ $V_2 = 9.2\ V$ $V_3 = 4.2\ V$



Electrical Circuit (1)

Introduction (week3 class2)

Dr. Akram Al-Mahrouk

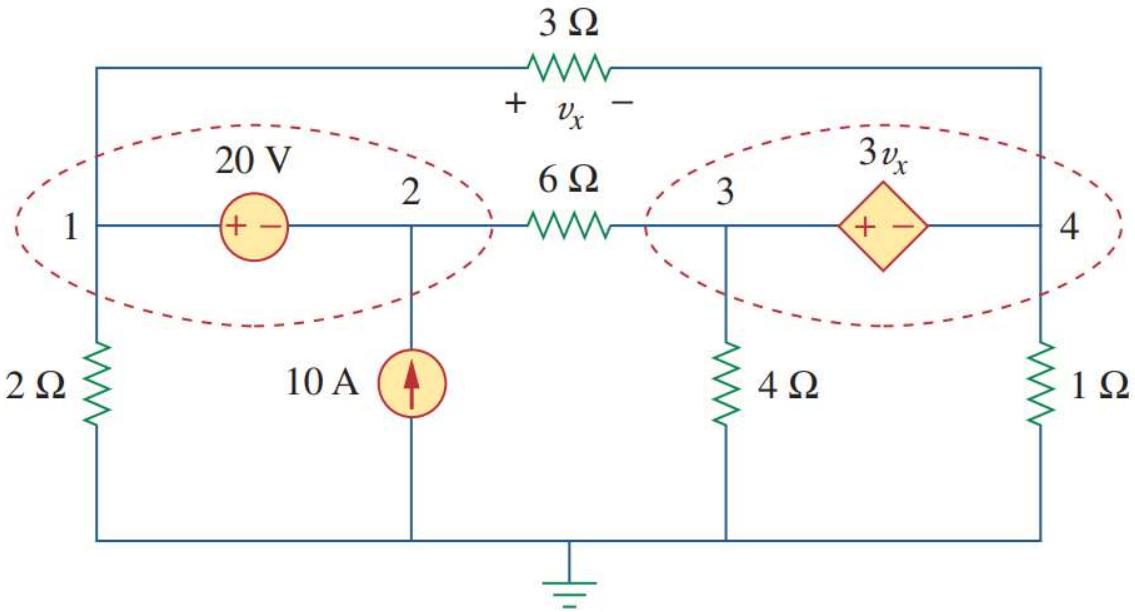
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Chapter 3 Methods of Analysis 81

3.1 Introduction 82
3.2 Nodal Analysis 82
3.3 Nodal Analysis with Voltage Sources 88
3.4 Mesh Analysis 93
3.5 Mesh Analysis with Current Sources 98
3.6 †Nodal and Mesh Analyses by Inspection 100
3.7 Nodal Versus Mesh Analysis 104
3.8 Circuit Analysis with *PSpice* 105
3.9 †Applications: DC Transistor Circuits 107
3.10 Summary 112
Review Questions 113
Problems 114
Comprehensive Problem 126

Advanced

Nodal Analysis with Voltage Sources



Nodal Analysis with Voltage Sources

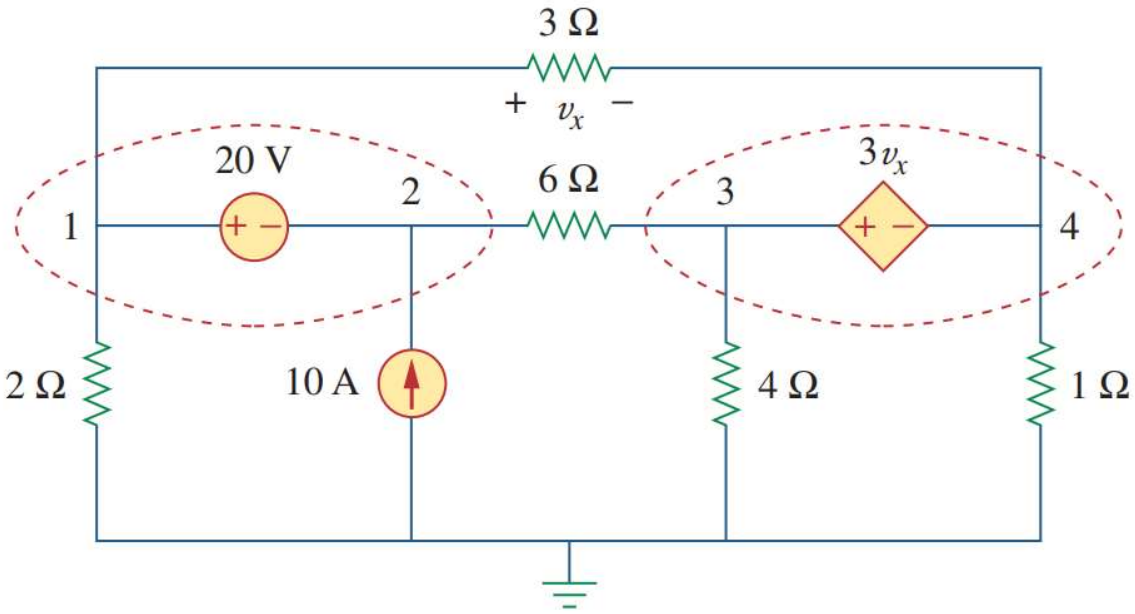
Basic equation

$$V_1 = V_2 + 20$$

$$V_3 = V_4 + 3(V_1 - V_4)$$

$$\frac{V_1}{2} + \frac{V_1 - V_4}{3} + (-10) + \frac{V_2 - V_3}{6} = 0$$

$$\frac{V_3}{4} + \frac{V_3 - V_2}{6} + \frac{V_4}{1} + \frac{V_4 - V_1}{3} = 0$$



Formulated equation

$$V_1 - V_2 = 20$$

$$3V_1 - V_3 - 2V_4 = 0$$

$$\left(\frac{1}{2} + \frac{1}{3}\right)V_1 + \left(\frac{1}{6}\right)V_2 + \left(\frac{-1}{6}\right)V_3 + \left(\frac{-1}{3}\right)V_4 = 10$$

$$\frac{-1}{3}V_1 + \frac{-1}{6}V_2 + (\frac{1}{4} + \frac{1}{6})V_3 + (1 + \frac{1}{3})V_4 = 0$$

Final answer

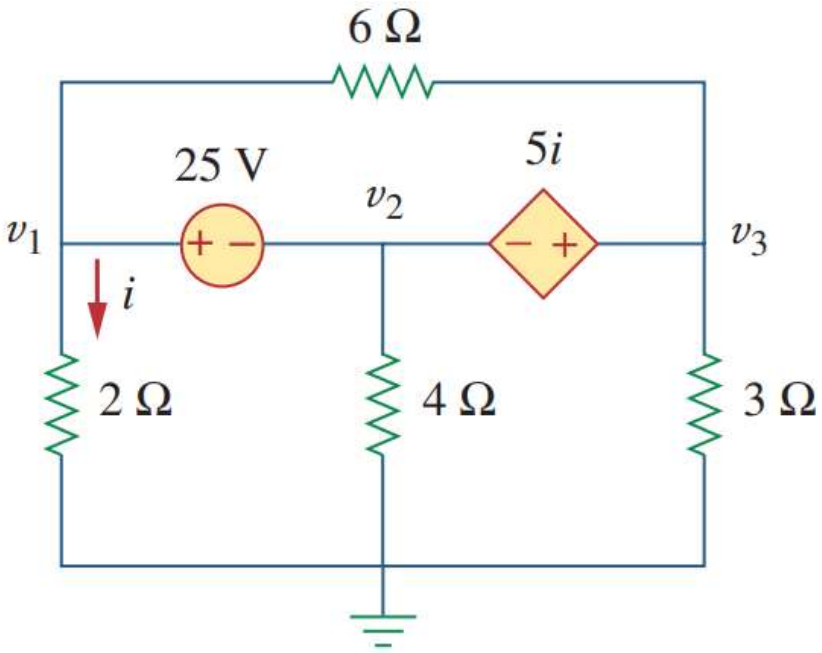
$$V_1 = -26.6667\ V$$

$$V_2 = 6.66667\ V$$

$$V_3 = 173.333\ V$$

$$V_4 = -46.6667\ V$$

Nodal Analysis with Voltage Sources



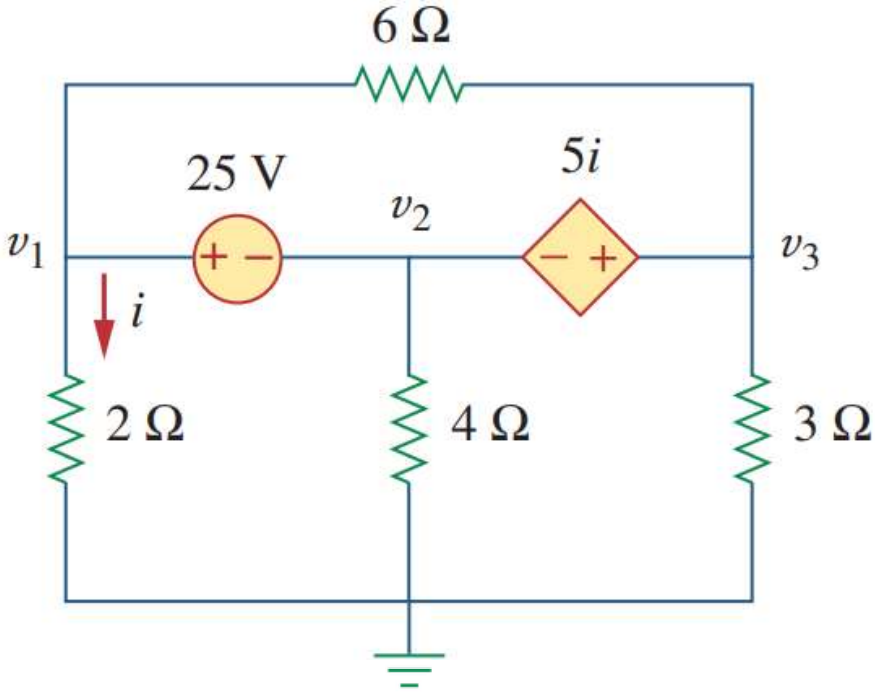
Nodal Analysis with Voltage Sources

Basic equation

$V_1 = V_2 + 25$

$V_3 = V_2 + 5(\frac{V_1}{2})$

$\frac{V_1}{2} + \frac{V_2}{4} + \frac{V_3}{3} = 0$



Formulated equation

$V_1 - V_2 = 25$

$(\frac{5}{2})V_1 + V_2 - V_3 = 0$

$\left(\frac{1}{2}\right)V_1 + \left(\frac{1}{4}\right)V_2 + \left(\frac{1}{3}\right)V_3 = 0$

Final answer

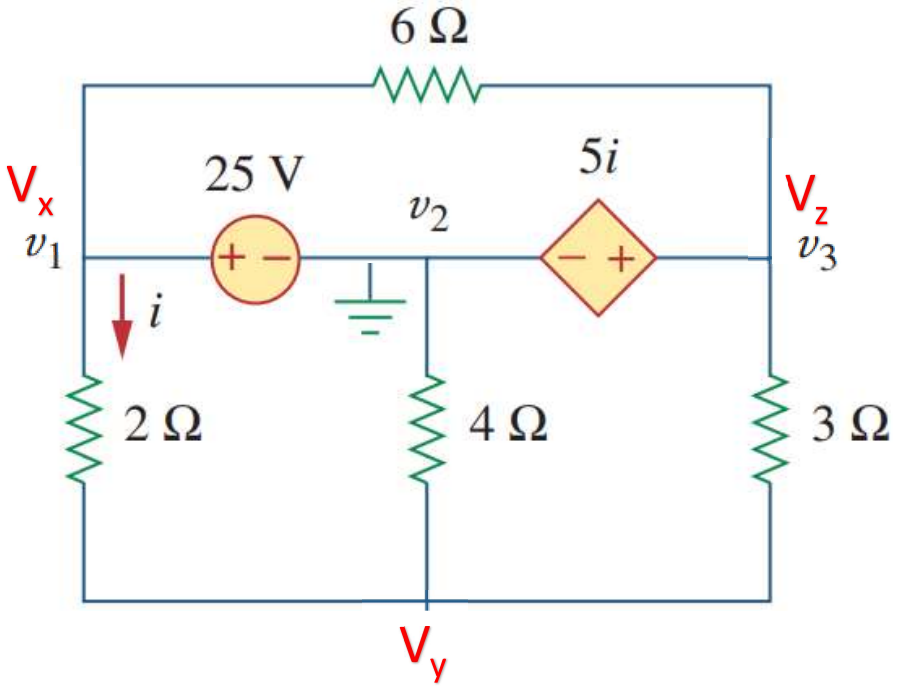
$V_1 = 7.608\ V$

$V_2 = -17.39\ V$

$V_3 = 1.630\ V$

Nodal Analysis with Voltage Sources

Without super node



Nodal Analysis with Voltage Sources

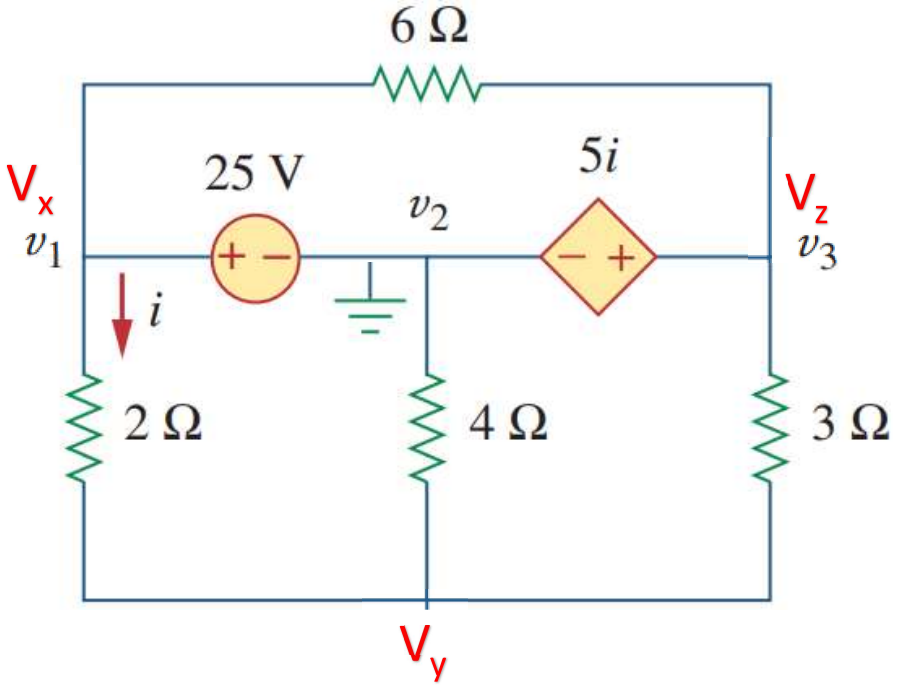
Without super node

Basic equation

$V_x = 25$

$V_z = 5(\frac{V_x - V_y}{2})$ ⇒ $V_z = 5(\frac{25 - V_y}{2})$

$\frac{V_y - V_x}{2} + \frac{V_y}{4} + \frac{V_y - V_z}{3} = 0$ ⇒ $\frac{V_y - 25}{2} + \frac{V_y}{4} + \frac{V_y - V_z}{3} = 0$



Formulated equation

$\left(\frac{5}{2}\right)V_y + V_z = \left(\frac{5 * 25}{2}\right)$

$\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{3}\right)V_y + \frac{-1}{3}V_z = \frac{25}{2}$

Final answer

$V_x = 25\ V$

$V_y = 17.39\ V$

$V_z = 19.021\ V$

$V_1 = V_x - V_y = 25 - 17.39 = 7.608\ V$

$V_2 = -V_y = -17.39\ V$

$V_3 = V_z - V_y = 19.021 - 17.39 = 1.630\ V$



Electrical Circuit (1)

Introduction (week5 class1)

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Chapter 3 Methods of Analysis 81

3.1 Introduction 82
3.2 Nodal Analysis 82
3.3 Nodal Analysis with Voltage Sources 88
3.4 Mesh Analysis 93
3.5 Mesh Analysis with Current Sources 98
3.6 †Nodal and Mesh Analyses by Inspection 100
3.7 Nodal Versus Mesh Analysis 104
3.8 Circuit Analysis with *PSpice* 105
3.9 †Applications: DC Transistor Circuits 107
3.10 Summary 112
Review Questions 113
Problems 114
Comprehensive Problem 126

Mesh Analysis

Steps to Determine Mesh Currents:

1. Assign mesh currents $i_1, i_2, \ldots, i_n$ to the n meshes.
2. Apply KVL to each of the n meshes. Use Ohm's law to express the voltages in terms of the mesh currents.
3. Solve the resulting n simultaneous equations to get the mesh currents.

Mesh Analysis

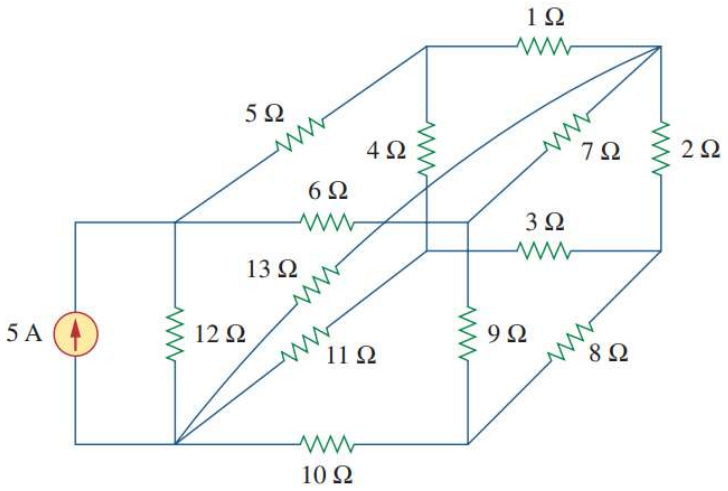
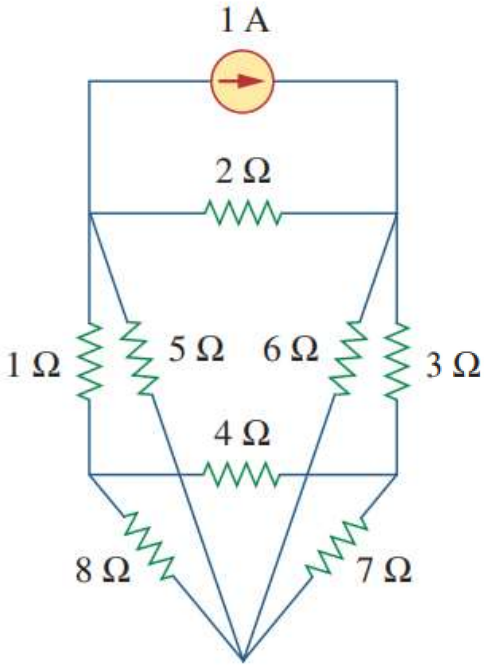
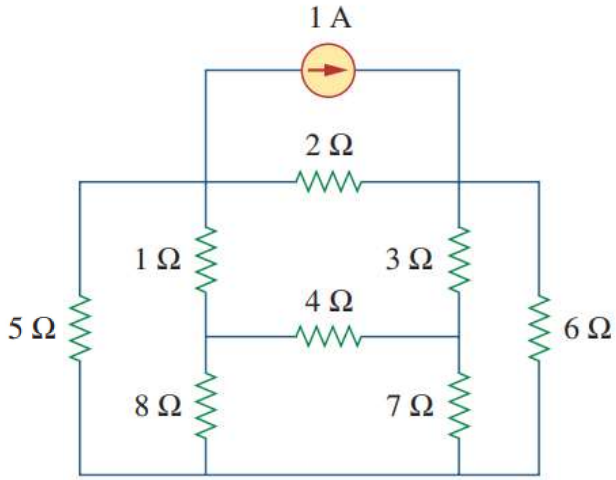


Figure 3.16
A nonplanar circuit.



=



Mesh Analysis

Basic equation

$$-15 + (5 * i_1) + 10 * (i_1 - i_2) + 10 = 0$$

$$-10 + 10 * (i_2 - i_1) + (6 * i_2) + (4 * i_2) = 0$$

Formulated equation

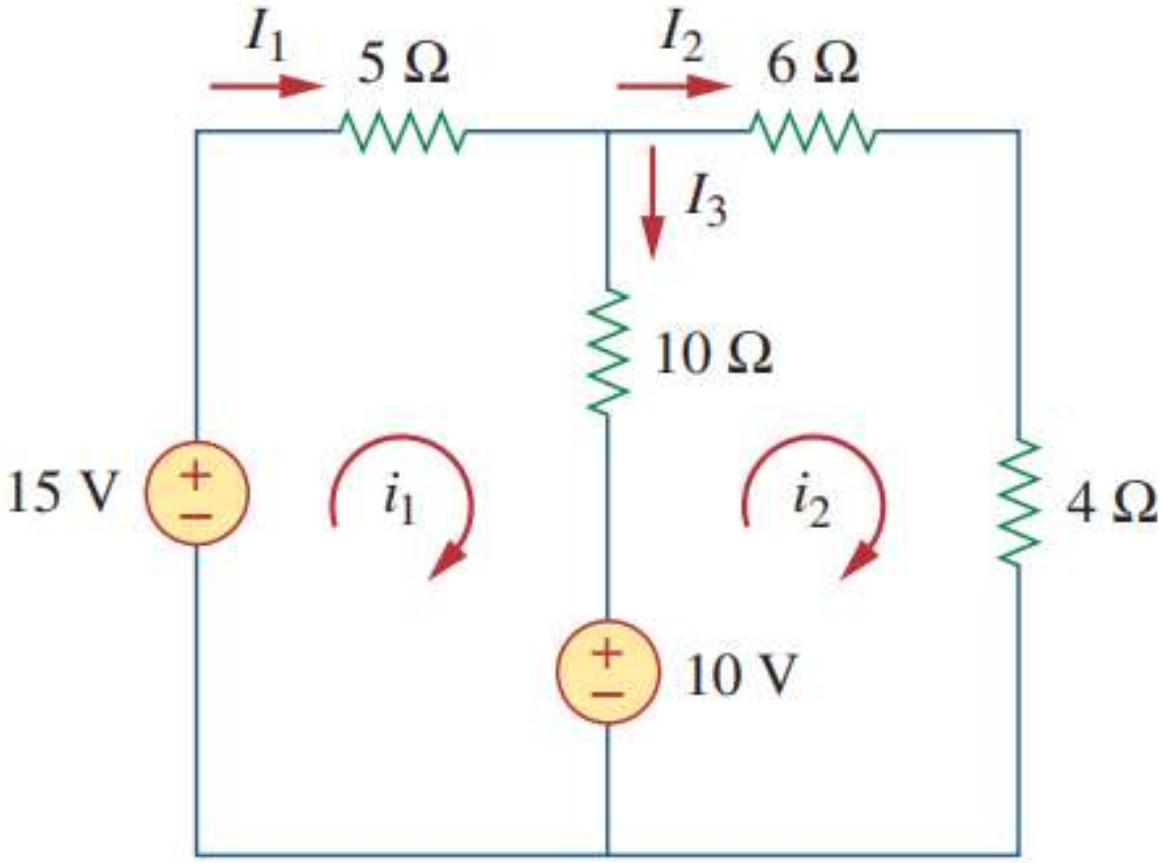
$$15i_1 - 10i_2 = 5$$

$$-10i_1 + 20i_2 = 10$$

Final answer

$$i_1 = 1\ \text{A}$$

$$i_2 = 1\ \text{A}$$



Mesh Analysis

Basic equation

$$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0$$

$$10(i_2 - i_1) + 24i_2 + 4(i_2 - i_3) = 0$$

$$12(i_3 - i_1) + 4(i_3 - i_2) + 4(i_1 - i_2) = 0$$

Formulated equation

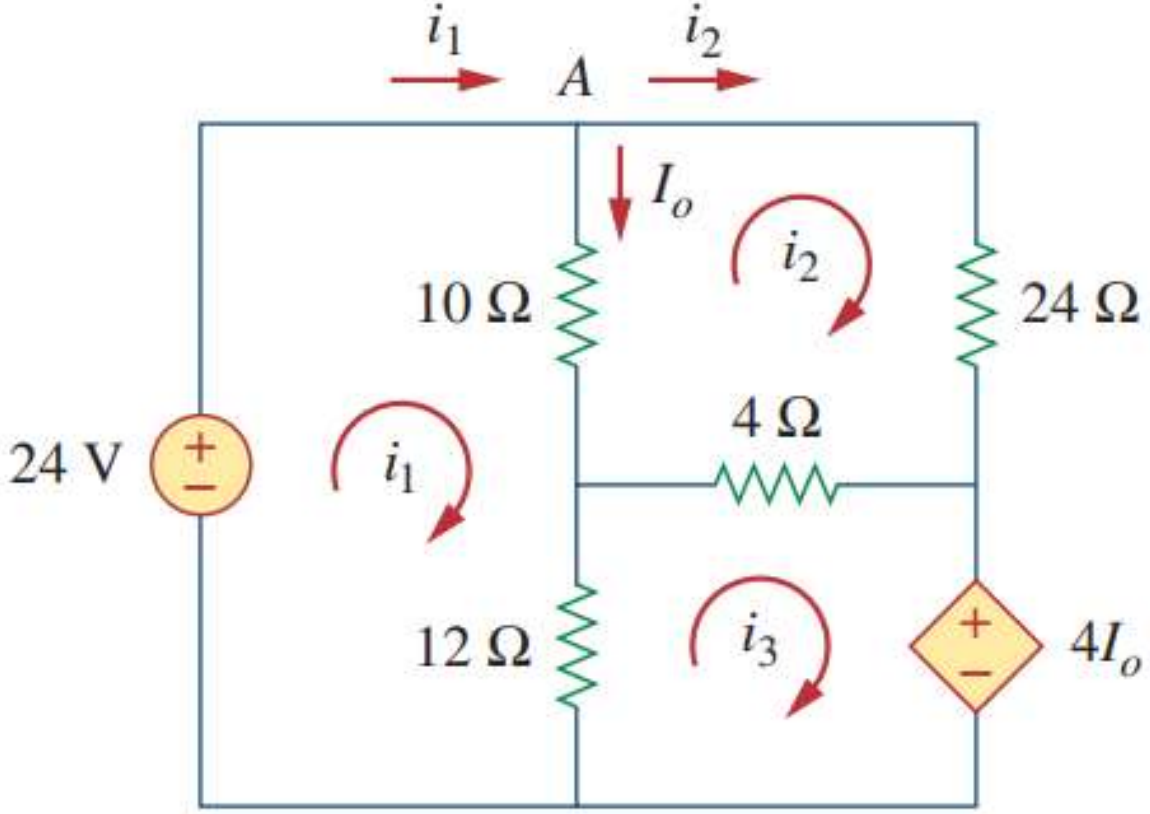
$$22i_1 - 10i_2 - 12i_3 = 24$$

$$-10i_1 + 38i_2 - 4i_3 = 0$$

$$-8i_1 - 8i_2 + 16i_3 = 0$$

Final answer

$i_1 = 2.25A$ $i_2 = 0.75\ A$ $i_3 = 1.5\ A$





Electrical Circuit (1)

Introduction (week5 class2)

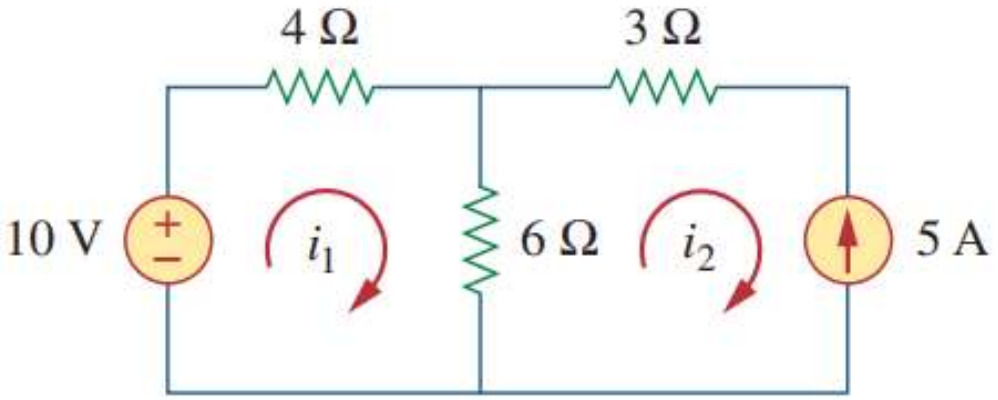
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Chapter 3 Methods of Analysis 81

3.1 Introduction 82
3.2 Nodal Analysis 82
3.3 Nodal Analysis with Voltage Sources 88
3.4 Mesh Analysis 93
3.5 Mesh Analysis with Current Sources 98
3.6 †Nodal and Mesh Analyses by Inspection 100
3.7 Nodal Versus Mesh Analysis 104
3.8 Circuit Analysis with *PSpice* 105
3.9 †Applications: DC Transistor Circuits 107
3.10 Summary 112
Review Questions 113
Problems 114
Comprehensive Problem 126

Mesh Analysis with Current Sources

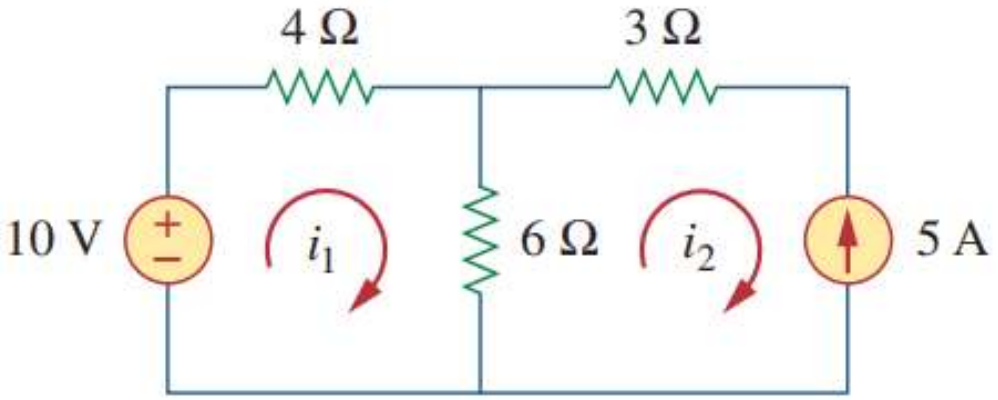


Mesh Analysis with Current Sources

Basic equation

$$-10 + (4 * i_1) + 6 * (i_1 - i_2) = 0$$

$$i_2 = -5$$

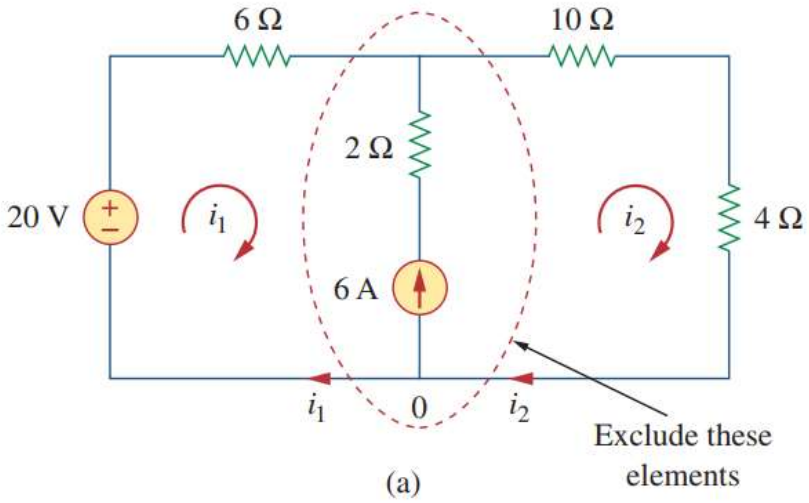


Final answer

$$i_1 = -2 \text{ A}$$

$$i_2 = -5 \text{ A}$$

Mesh Analysis with Current Sources



Mesh Analysis with Current Sources

Basic equation

$i_1 + 6 = i_2$

$-20 + (6 * i_1) + (10 * i_2) + (4 * i_2) = 0$

Formulated equation

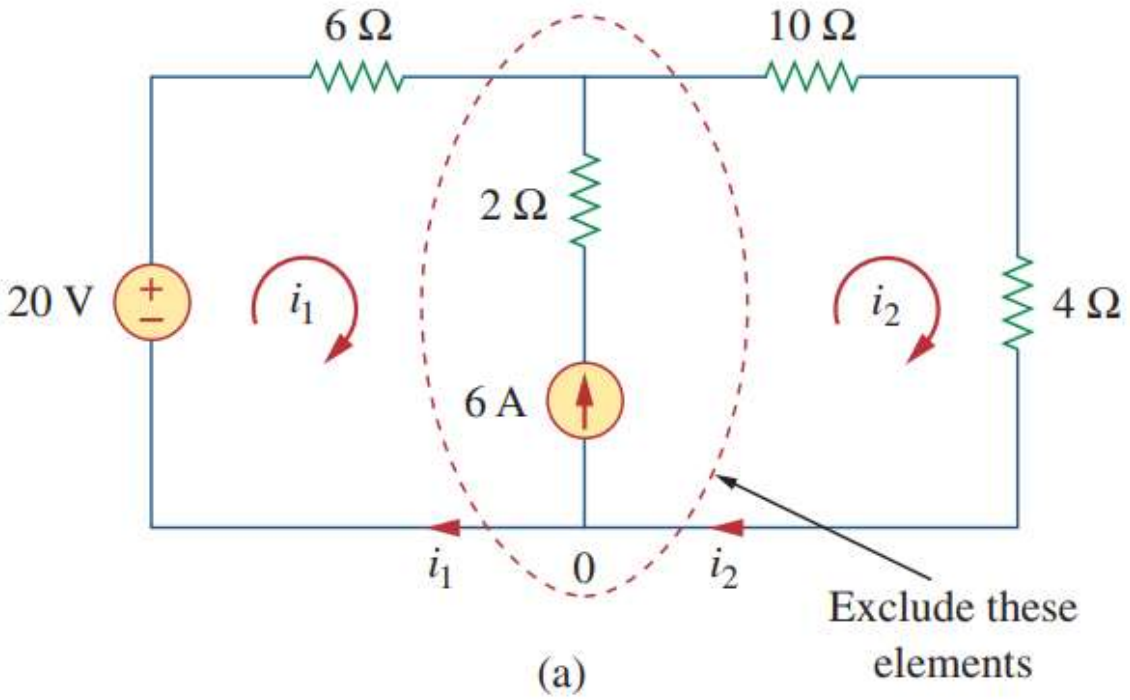
$i_1 - i_2 = -6$

$6i_1 + 14i_2 = 20$

Final answer

$i_1 = -3.2$ A

$i_2 = 2.8$ A





Electrical Circuit (1)

Circuit Technique (week6 class1)

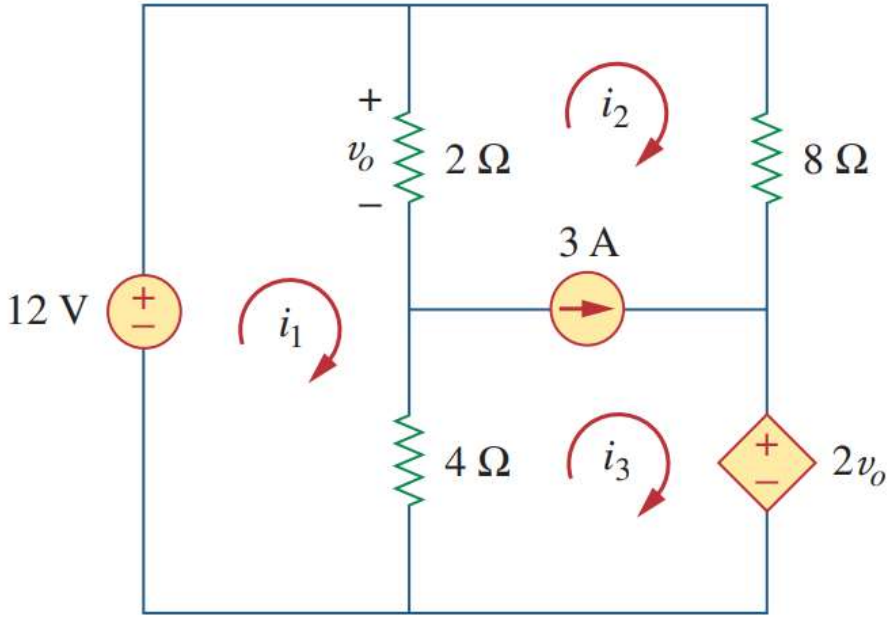
Dr. Akram Al-Mahrouk

Philadelphia University

Chapter 3 Methods of Analysis 81

3.1 Introduction 82
3.2 Nodal Analysis 82
3.3 Nodal Analysis with Voltage Sources 88
3.4 Mesh Analysis 93
3.5 Mesh Analysis with Current Sources 98
3.6 †Nodal and Mesh Analyses by Inspection 100
3.7 Nodal Versus Mesh Analysis 104
3.8 Circuit Analysis with *PSpice* 105
3.9 †Applications: DC Transistor Circuits 107
3.10 Summary 112
Review Questions 113
Problems 114
Comprehensive Problem 126

Mesh Analysis with Current Sources



Mesh Analysis with Current Sources

Basic equation

$$-12 + 2(i_1 - i_2) + 4(i_1 - i_3) = 0$$

$$i_3 - i_2 = 3$$

$$4(i_3 - i_1) + 2(i_2 - i_1) + 8i_2 + 4(i_1 - i_2)) = 0$$

Formulated equation

$$6i_1 - 2i_2 - 4i_3 = 12$$

$$-i_2 + i_3 = 3$$

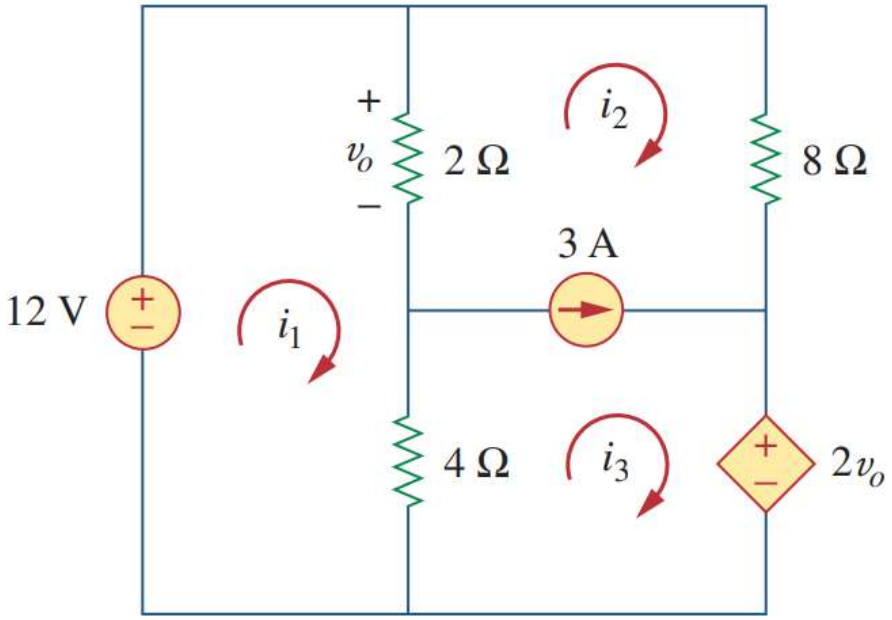
$$-2i_1 + 6i_2 + 4i_3 = 0$$

Final answer

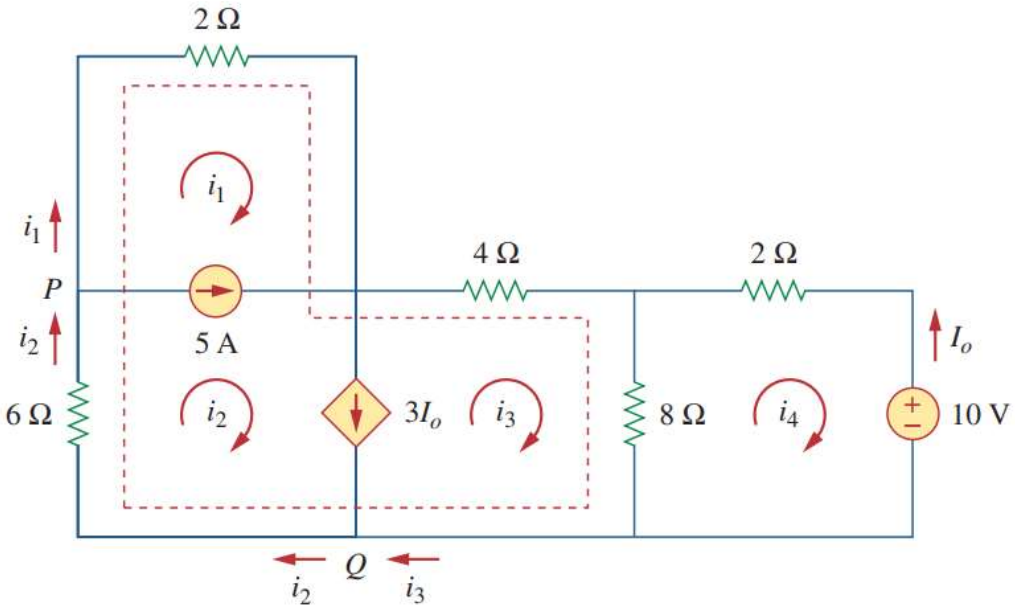
$$i_1 = 3.5 \text{ A}$$

$$i_2 = -0.5 \text{ A}$$

$$i_3 = 2.5 \text{ A}$$



Mesh Analysis with Current Sources



Mesh Analysis with Current Sources

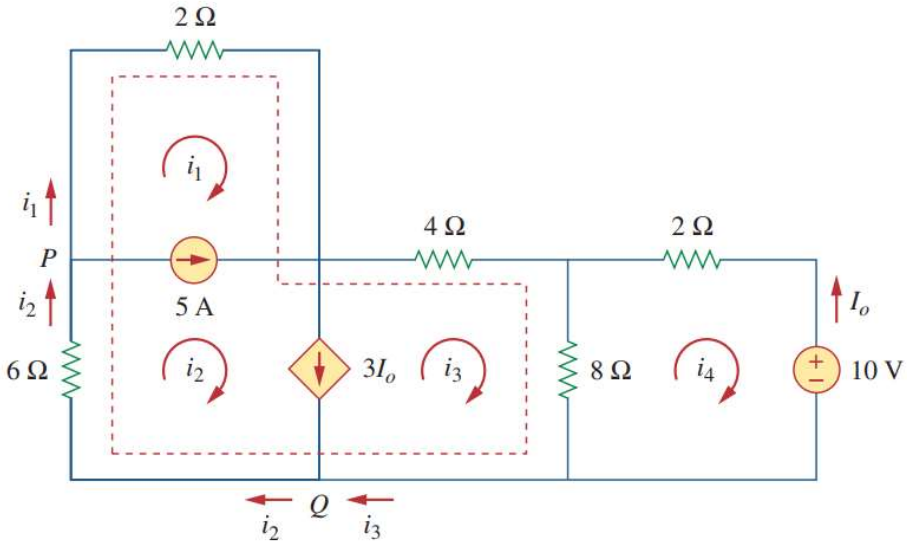
Basic equation

$i_2 - i_1 = 5$

$i_2 - i_3 = 3i_0 \qquad i_0 = -i_4$

$6i_2 + 2i_1 + 4i_3 + 8(i_3 - i_4) = 0$

$8(i_4 - i_3) + 2i_4 + 10 = 0$



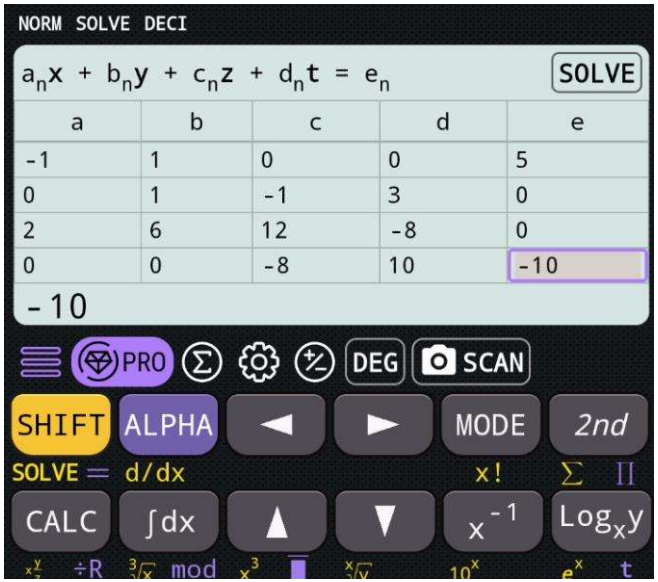
Formulated equation

$-i_1 + i_2 = 5$

$i_2 - i_3 + 3i_4 = 0$

$2i_1 + 6i_2 + 12i_3 - 8i_4 = 0$

$-8i_3 + 10i_4 = -10$



Final answer

$i_1 = -7.5$ A

$i_2 = -2.5$ A

$i_3 = 3.9286$ A

$i_4 = 2.1429$ A

